MSBA 265: Business Analytics Topics (3 Units)

Eberhardt School of Business — University of the Pacific

Day 1 Course Orientation, Curriculum Architecture, & Syllabus

Instructor: Visiting Instructor – Shyla Solis (`ssolis1@pacific.edu`)

Term: FALL 2026 • **Schedule:** MW 6:30–7:45 PM • **Location:** Weber 107

Welcome to MSBA 265 at Eberhardt!

"In this course, you will learn to transform raw, chaotic data into production-grade predictive models, containerized microservices, and actionable business intelligence for modern enterprise leadership."

Contents

1 Administrative Information & Course Overview

Table 1: Course Quick-Reference Metadata

Institution	University of the Pacific, Eberhardt School of Business
Course Code	MSBA 265 – Business Analytics Topics (3 Units)
Term & Location	Fall 2026 | Weber Hall 107 | MW 6:30–7:45 PM
Instructor	Visiting Instructor – Shyla Solis
Contact & Office Hours	Email: `ssolis1@pacific.edu` | Office Hours: By appointment ONLY
Prerequisites	Graduate standing in the MSBA program or instructor approval. Prior exposure to statistics and basic programming recommended.

1.1 Course Description

This course familiarizes graduate students with a broad cross-section of models and algorithms used in machine learning and business analytics. Students will learn to make sense of data using data cleaning techniques, exploratory data analysis, visualization, and data mining algorithms applied to real-world datasets relevant to modern business problems. The course also introduces the fundamentals of Natural Language Processing (NLP), enabling students to perform text processing and modeling on document and social media data.

1.2 Course Learning Objectives (CLO)

By the end of this course, students will be able to:

CLO 1: Clean, preprocess, and explore structured and unstructured business datasets.

CLO 2: Apply supervised and unsupervised machine learning models to enterprise problems.

CLO 3: Evaluate and compare predictive models using appropriate performance metrics.

CLO 4: Perform text preprocessing and basic NLP modeling on real-world text data.

CLO 5: Communicate analytical findings effectively to both technical and non-technical audiences.

CLO 6: Translate data-driven insights into actionable executive business recommendations.

1.3 Required Software & Environment Stack

Students are expected to bring a laptop to every class session capable of executing:

- **Language Stack:** Python 3.10+ (Anaconda Distribution recommended).
- **Development Environment:** Jupyter Notebook / JupyterLab.
- **Core Analytics Libraries:** Pandas, NumPy, scikit-learn.
- **Visualization Tools:** Matplotlib, Seaborn, or Plotly.
- **NLP Frameworks:** NLTK and/or spaCy (for text mining modules).
- **Deployment Stack:** FastAPI, Pydantic, joblib, and Docker Desktop.

2 Visual Course Roadmap & Learning Sequence

Figure 1 maps out your journey across the Spring 2026 term. Notice how the **Four Foundational Modules** establish the engineering baseline that powers every applied project. Each project is followed directly by an exam to test individual mastery, leading into your final Enterprise Capstone System.

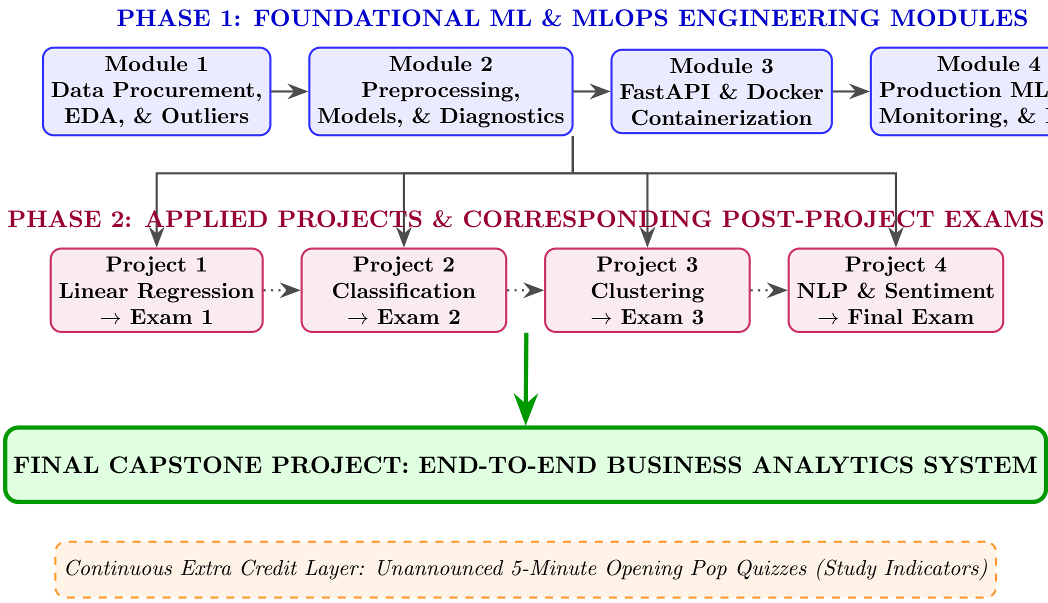

Figure 1: MSBA 265 Course Architecture: Modules, Applied Projects, Sequential Exams, and Capstone System.

3 Tentative Schedule & Topic Alignment

Table 2 outlines the alignment between the foundational course modules, core analytics topics, practical projects, and subsequent exams across the term.

**Note: The instructor reserves the right to adjust the schedule and topics as needed to optimize learning outcomes.*

4 Assessment Framework & Grading Policy

Grades in MSBA 265 are calculated across four primary evaluated categories, supported by an unannounced opening pop quiz extra credit mechanism:

4.1 Exam Structure & AI Disclosure

Modern business analytics tools and Generative AI assistants can aid in writing project code. However, **your true individual understanding of business analytics concepts will be reflected on your exams.**

To ensure fair and rigorous evaluation, all exams (Exams 1, 2, 3, and the Final Exam) will follow a dual format:

1. **Multiple Choice Questions:** Testing core algorithm mechanics, diagnostic criteria, metric selection, and pipeline logic.

2. **Open-Ended Writing Questions:** Requiring you to interpret model output, explain trade-offs (e.g., precision vs. recall, X vs. Y transformations), and translate technical metrics into executive business recommendations.

Hands-on project work directly prepares you for these exams. If you actively complete and understand your project implementations, you will be well-prepared for the exam questions.

4.2 Unannounced Opening Pop Quizzes (Extra Credit)

To help you gauge what will appear on upcoming exams and identify key study topics, **unannounced pop quizzes** will be handed out at the exact start of class (3:30 PM).

- **Timing & Duration:** Quizzes run strictly for the **first 5 minutes** of class.
- **Extra Credit Value:** Quizzes are graded as **Extra Credit** (adding up to +3.0 percentage points directly onto your final course grade).
- **Strict No-Make-Up Policy:** If you arrive late or are absent for any reason when a pop quiz is distributed, **you cannot make up the quiz**. No exceptions will be granted.

4.3 Grade Breakdown Scale

Final letter grades will be assigned according to the standard graduate scale:

Grade	Percentage Range	Grade	Percentage Range	Grade	Percentage Range
A	93.0% – 100%	B+	87.0% – 89.9%	C+	77.0% – 79.9%
A-	90.0% – 92.9%	B	83.0% – 86.9%	C	73.0% – 76.9%
		B-	80.0% – 82.9%	F	< 73.0%

5 Attendance Policy, Discrete Deductions, & Recovery Mechanism

5.1 Attendance Policy & Discrete Absence Penalty

Active in-class engagement is a vital component of a graduate analytics program. Attendance will be recorded at the start of every session (TTH 3:30–4:45 PM).

DISCRETE ABSENCE PENALTY RULE:
Each unexcused absence results in a ***discrete deduction of 3.0 percentage points (**3.0%**) directly from your final numerical course grade****. On the exact 10 minute mark after class begins, counts as a late arrival; two late arrivals equal one complete absence (penalty will be applied).*

5.1.1 Why This Rule Exists (How the Math Works)

Life happens—flat tires, sudden illnesses, or unexpected delays occur. The 3.0-point deduction is mathematically structured so that an occasional emergency will not ruin an otherwise strong grade, while repeated absences will significantly impact your final letter grade:

- **1 Absence (−3.0%):** *Mild Impact.* If your course average is a 95% (A), one absence drops your grade to a 92% (A−). You are still in good standing.
- **3 Absences (−9.0%):** *Major Impact.* Drops an A student (95%) down to an 86% (B).
- **4 Absences (−12.0%):** *Severe Impact.* Drops an A student (95%) down to an 83% (B−), pushing a B student down to a C.

5.2 Absence Point Recovery via LinkedIn Learning

If you miss a class, you have the opportunity to recover your deducted 3.0% per absence by utilizing the **University of the Pacific's institutional LinkedIn Learning benefit**:

1. **Identify Missed Topic:** Check the course schedule or lecture slides for the specific date and topic you missed.
2. **Complete Approved Course Module:** Log into LinkedIn Learning using your Pacific SSO credentials. Complete an instructor-approved course module that directly matches the technical topic covered on the day of your absence (e.g., *"Exploratory Data Analysis in Python"* or *"Building REST APIs with FastAPI"*).
3. **Submit your completion certificate:** Email your completed **LinkedIn learning certificate** to Instructor Solis (`ssolis1@pacific.edu`) within **7 calendar days** of missed class.
4. **Point Restoration:** Upon verification, the 3.0% deduction for that specific absence will be fully restored to your final grade.
5. **Student Communication:** All student communications, whether by email, verbal, or otherwise, will be documented in the student communications sheet. This is an excel sheet to keep track of commitments made between students and the instructor.

5.3 General Course Policies & Standards

1. **The Anti-Sanitization Mandate:** Using pre-cleaned educational datasets (such as Iris, Titanic, or Boston Housing) for any module or project is strictly prohibited. Submitting a pre-sanitized dataset will result in an automatic score of zero for that assignment.

2. **End-to-End Git, Pipeline, & Container Reproducibility Standard:** In the software industry, code that only runs on a local laptop is considered broken. Every project must live in a dedicated Git repository (e.g., GitHub) and be fully executable from the Command Line / Terminal. Anyone in the world reading your `README.md` must be able to clone your repository, follow your instructions, and reproduce your exact results end-to-end. Your documentation must include explicit, tested setup steps for **both macOS and Windows** environments, utilizing scikit-learn `Pipeline` objects for atomic model bundling and/or `Docker` containers for environment isolation. Instructions must also detail how to programmatically extract or download the raw online dataset directly into the pipeline. If the teaching staff cannot clone and execute your project end-to-end from the terminal using your Git instructions, the submission will be deemed non-reproducible and will receive a score of zero.

3. **Standard Project Repository Architecture Example:** To satisfy the reproducibility standard, your Git repository should follow this professional software layout:

```
my-msba265-project/
|-- .gitignore                    # Excludes raw data, OS files, or local cache
|-- Dockerfile                    # Container setup (guarantees Mac/Win parity)
|-- README.md                       # STEP-BY-STEP CLI instructions for Mac/Win
|-- requirements.txt                        # Exact Python package dependencies
|
|-- data/
|    `-- download_data.py                        # Script to download raw dataset
|
|-- src/
|    |-- preprocess.py                         # Data cleaning and transformer code
|    |-- train.py                      # Fits scikit-learn Pipeline & saves .joblib
|    `-- main.py                             # FastAPI microservice loading .joblib
|
`-- tests/
`-- test_api.py                                # CLI tests verifying API endpoints
```

4. **Late Submission Policy:** Projects lose 10% of total value per 24-hour period past the deadline. Submissions made more than 72 hours late will not be accepted.

5. **Office Hours:** Office hours with Instructor Solis are conducted **by appointment ONLY**. Requests must be sent via email (`ssolis1@pacific.edu`) at least 24 hours in advance.

6. **Academic Integrity & Generative AI:** You are permitted to use AI tools as coding assistants during projects. However, submitting work you cannot explain during exams or misrepresenting AI-generated output as original analytical thought violates the University of the Pacific Honor Code.

Table 2: Spring 2026 MSBA 265 Tentative Schedule & Module Map

Term Phase	Course Topic Modules	Deliverables & Milestone Exams
Weeks 1–2	**Module 1: Foundations & Procurement** Analytics lifecycle, ethics, raw data procurement, univariate EDA, skewness, and $1.5 \times$ IQR outlier filters.	Foundational Assignment 1 (Procurement Audit)
Weeks 3–4	**Module 2: Data Preprocessing & Cleaning** Pearson correlation matrix, multicollinearity ($\|r\| \geq 0.85$), Z-score scaling, dummy encoding, and residual checks.	Foundational Assignment 2 (Pipeline & Diagnostics)
Weeks 5–6	**Module 3: Continuous Regression & Deployment** Linear models, regularized loss (Lasso/Ridge/ElasticNet), model serialization, FastAPI, and Docker packaging.	**Project 1 Due** $\rightarrow$ **Exam 1 (Post-Project 1)**
Weeks 7–8	**Module 4: Supervised Classification** Logistic regression, decision trees, k-NN, ROC-AUC, confusion matrices, and class balance mitigation.	**Project 2 Due** $\rightarrow$ **Exam 2 (Post-Project 2)**
Weeks 9–11	**Module 5: Unsupervised Learning & Clustering** Clustering (k-Means, hierarchical), customer segmentation, anomaly detection, and PCA dimension reduction.	**Project 3 Due** $\rightarrow$ **Exam 3 (Post-Project 3)**
Weeks 12–13	**Module 6: Natural Language Processing (NLP)** Text preprocessing, tokenization, TF-IDF, sentiment analysis, document classification using NLTK/spaCy.	**Project 4 Due** $\rightarrow$ **Final Exam (Cumulative)**
Weeks 14–15	**Module 7: Applied MLOps & Final Capstone** Concept drift monitoring, model explainability (SHAP), enterprise reporting, and Capstone presentations.	**Final Capstone System & Executive Presentation**

Table 3: Course Assessment Distribution Matrix

Assessment Component	**Weight**	**Description & Structure**
Exams (1, 2, 3, & Final)	35%	**Multiple Choice + Open-Ended Writing.** Conducted in-class following each project. Evaluates real individual knowledge retention.
Applied Projects (1–4)	35%	Hands-on ML system builds (Linear Regression, Classification, Clustering, NLP). Projects serve as direct study vehicles for exams.
Final Project	20%	End-to-end production-grade enterprise system with an executive presentation and technical runbook.
Foundational Modules (1–4)	10%	Hands-on assignments establishing data engineering, scaling, FastAPI, and MLOps baselines.
Opening Pop Quizzes	**Extra Credit**	Unannounced 5-minute quizzes randomly given at the start of class. Adds up to **+3.0%** to final grade. Serves as direct exam preparation.